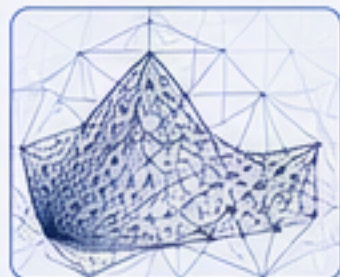


1 NONLINEAR MATHEMATICS & NUMERICAL ANALYSIS

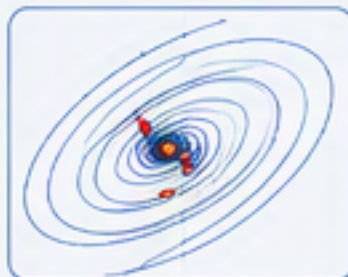
$$\frac{\partial u}{\partial t} = N(u)$$

$$\nabla \cdot (k \nabla u) = f$$

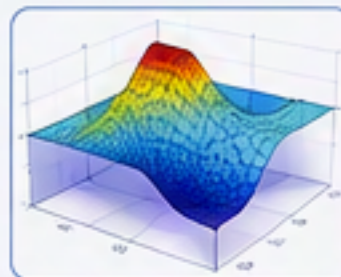
- Nonlinear differential equations
- Numerical analysis
- Adaptive algorithms
- Scientific computing



Adaptive meshes



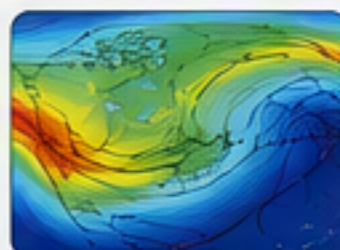
Dynamical systems



Scientific computing

2 WEATHER & CLIMATE

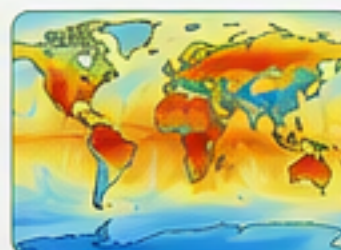
- Numerical weather prediction
- Data assimilation
- Met Office collaboration
- Climate modelling
- Sustainability science



Global weather models



Data assimilation



Climate projections

CHRIS BUDD OBE

Mathematics for a Complex World

Professor of Applied Mathematics

Department of Mathematical Sciences, University of Bath

I use mathematics, analysis and computation to understand complex nonlinear systems and to solve real-world problems that matter – from weather and climate to industry and society.

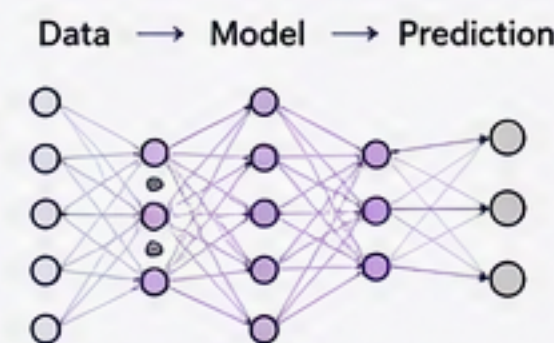


$$\frac{\partial u}{\partial t} = N(u, \nabla u, \nabla^2 u, \dots)$$

Solve → Understand → Impact

3 SCIENTIFIC MACHINE LEARNING

- Scientific machine learning
- Mathematics for deep learning
- AI guided by physical laws
- Hybrid modelling
- Inverse problems

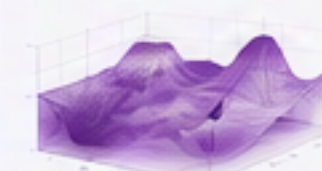


PDEs & Physics

$$\frac{\partial u}{\partial t} = N(u)$$

$$\mathcal{L}(u) = 0$$

Neural Operators & PINNs



Uncertainty Quantification



4 INDUSTRIAL MATHEMATICS & REAL-WORLD APPLICATIONS

- Industrial collaboration
- Applied mathematics
- Mathematical innovation
- Real-world impact



Aerospace



Energy



Energy Systems



Telecommunications



Food & Agriculture



Other Industries

5 COMPLEXITY & DYNAMICAL SYSTEMS



Chaos & Strange Attractors



Complex Networks



Tipping Points & Thresholds



Uncertainty Quantification

- Emergent behaviour
- Nonlinear dynamics
- Multi-scale phenomena
- Uncertainty & risk
- Complex systems theory

IMPACT: FROM MATHEMATICS TO SOCIETY



Weather forecasting



Climate policy



Energy systems



Aerospace engineering



Telecoms networks

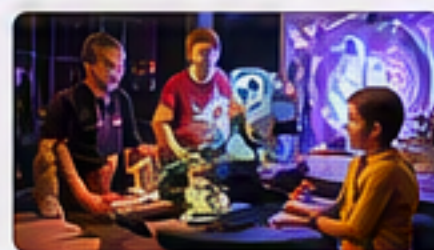


Food production

PUBLIC ENGAGEMENT & EDUCATION



Professor of Mathematics
Royal Institution



Bath Taps Into Science
Science Festival Director



Inspiring the next generation
Public talks & schools outreach

Public understanding of mathematics

Citizen science projects

Communicating the excitement of maths and science

LEADERSHIP & COLLABORATION

University of Bath
Professor of Applied Mathematics

CliMathNet
Co-Director
(Climate Mathematics Network)

Institute for Mathematical
Innovation (IMI)
Director of Knowledge Exchange

Interdisciplinary collaborations
nationally and internationally



Working with universities, research institutes,
industry, government and policymakers worldwide